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Cooling device

Abstract

A cooling device includes a heat-receiving block on which a heat emitter is mounted, a first heat pipe fixed at the heat-receiving block, and a second heat pipe located adjacent to the first heat pipe. The first heat pipe accommodates a first refrigerant in the gas-liquid two-phase state, and the second heat pipe accommodates a second refrigerant in the gas-liquid two-phase state. The ratio of the amount of the second refrigerant in the liquid state to the volume of the second heat pipe is higher than the ratio of the amount of the first refrigerant in the liquid state to the volume of the first heat pipe.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a cooling device including heat pipes.

BACKGROUND ART

(2) In order to avoid damage caused by heat emission of electronic components during energization, the electronic components are thermally coupled to a cooling member. The cooling member discharges the heat transferred from the electronic components to the air around the cooling member. The electronic components are accordingly cooled. A typical example of the cooling member is a heat sink including heat pipes. This type of heat sink is disclosed in Patent Literature 1, for example. The heat sink disclosed in Patent Literature 1 includes a base plate to receive heat transferred from electronic components, and heat pipes. The heat pipes include a plate heat pipe fixed at the base plate and cylindrical heat pipes communicating with the plate heat pipe.

CITATION LIST

Patent Literature

(3) Patent Literature 1: Unexamined Japanese Patent Application Publication No. 2003-336976

SUMMARY OF INVENTION

Technical Problem

(4) The heat pipes included in the heat sink disclosed in Patent Literature 1 accommodate a refrigerant in the gas-liquid two-phase state. The refrigerant evaporated by the heat transferred from the electronic components flows from the plate heat pipe into the cylindrical heat pipes. While traveling inside the cylindrical heat pipes, the evaporated refrigerant transfers the heat via the cylindrical heat pipes to the air around the cylindrical heat pipes. The heat transfer from the refrigerant to the air lowers the temperature of the refrigerant and condenses the refrigerant into liquid. The refrigerant in the liquid state flows through the cylindrical heat pipes into the plate heat pipe. The refrigerant thus repeats evaporation and condensation and thereby circulates inside the heat pipes, resulting in cooling of the electronic components.

(5) This refrigerant may freeze when the heat sink is installed in a place in contact with the air at a temperature equal to or lower than the melting point of the refrigerant. In an exemplary case where the heat pipes accommodate pure water as the refrigerant and the heat sink is installed in a place in contact with the air at a temperature of 0° C. or less, the pure water enclosed in the heat pipes may freeze. The frozen refrigerant does not circulate inside the heat pipes and therefore impairs the cooling capacity of the heat sink. The electronic components, which have not been sufficiently cooled, may reach an excessively high temperature to break down.

(6) In response to the above issue, an objective of the present disclosure is to provide a cooling device capable of cooling electronic components even in a low-temperature environment.

Solution to Problem

(7) In order to achieve the above objective, a cooling device according to an aspect of the present disclosure includes: a heat-receiving block, a first heat pipe, a second heat pipe, a first refrigerant in the gas-liquid two-phase state, and a second refrigerant in the gas-liquid two-phase state. The heat-receiving block includes a first main surface to which a heat emitter is fixed. The first heat pipe is fixed at the heat-receiving block. The second heat pipe is fixed at the heat-receiving block and located adjacent to the first heat pipe. The first refrigerant is enclosed in the first heat pipe. The second refrigerant is enclosed in the second heat pipe. At normal temperature, the ratio of the amount of the second refrigerant in the liquid state to the volume of the second heat pipe is higher than the ratio of the amount of the first refrigerant in the liquid state to the volume of the first heat pipe.

Advantageous Effects of Invention

(8) The cooling device according to an aspect of the present disclosure includes the second heat pipe located adjacent to the first heat pipe. At normal temperature, the ratio of the amount of the second refrigerant in the liquid state to the volume of the second heat pipe is higher than the ratio

of the amount of the first refrigerant in the liquid state to the volume of the first heat pipe. The second heat pipe that less readily freeze than the first heat pipe is located adjacent to the first heat pipe, and therefore can rapidly melt the frozen first refrigerant inside the first heat pipe. The cooling device is therefore capable of cooling the heat emitter even in a low-temperature environment.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a cooling device according to Embodiment 1 of the present disclosure;
(2) FIG. **2** is a sectional view of the cooling device according to Embodiment 1 taken along the line A-A of FIG. **1**;
(3) FIG. **3** illustrates a second heat pipe according to Embodiment 1;
(4) FIG. **4** is a sectional view of a power conversion apparatus according to Embodiment 1;
(5) FIG. **5** is a sectional view of the power conversion apparatus according to Embodiment 1 taken along the line B-B of FIG. **4**;
(6) FIG. **6** is a perspective view of a cooling device according to Embodiment 2 of the present disclosure;
(7) FIG. **7** is a sectional view of the cooling device according to Embodiment 2 taken along the line C-C of FIG. **6**;
(8) FIG. **8** is a top view of a plate member constituting a second heat pipe according to Embodiment 2;
(9) FIG. **9** is a sectional view of a cooling device according to Embodiment 3 of the present disclosure;
(10) FIG. **10** is a top view of a plate member constituting a second heat pipe according to Embodiment 3;
(11) FIG. **11** is a sectional view of a cooling device according to Embodiment 4 of the present disclosure;
(12) FIG. **12** is a top view of a plate member constituting a second heat pipe according to Embodiment 4;
(13) FIG. **13** is a perspective view of a cooling device according to a modified embodiment; and
(14) FIG. **14** is a schematic diagram of a loop heat pipe according to another embodiment.
DESCRIPTION OF EMBODIMENTS
(15) A cooling device according to embodiments of the present disclosure is described in detail with reference to the accompanying drawings. The identical or corresponding components are provided with the same reference symbol.
Embodiment 1
(16) In order to avoid failure of electronic components due to heat emission from the electronic components during energization, the electronic components are thermally coupled to a cooling device for cooling the electronic components. A cooling device **1** according to Embodiment 1 illustrated in FIG. **1** includes a heat-receiving block **11** to which heat emitters are fixed, first heat pipes **12** fixed at the heat-receiving block **11** to discharge the heat transferred from the heat emitters and thereby cool the heat emitters, and second heat pipes **13** fixed at the heat-receiving block **11** and located adjacent to the first heat pipes **12**. Each of the first heat pipes **12** includes a primary pipe portion **12***a* fixed at the heat-receiving block **11**, and secondary pipe portions **12***b* that are communicating with the primary pipe portion **12***a* and extend in the direction away from the heat-receiving block **11**. The meaning of fixation encompasses integral formation. Specifically, the primary pipe portions **12***a* fixed at the heat-receiving block **11** may be formed integrally with the

heat-receiving block **11**. Also, the second heat pipes **13** fixed at the heat-receiving block **11** may be formed integrally with the heat-receiving block **11**.

(17) As illustrated in FIG. **2**, which is a sectional view taken along the line A-A of FIG. **1**, the cooling device **1** further includes fins **14** fixed across the secondary pipe portions **12***b*. The fins **14** are not shown in FIG. **1** in order to facilitate an understanding. The cooling device **1** also includes a first refrigerant **15** in the gas-liquid two-phase state enclosed in the first heat pipes **12**, and a second refrigerant **16** in the gas-liquid two-phase state enclosed in the second heat pipes **13**.

(18) In FIGS. **1** and **2**, the Z axis corresponds to the vertical direction. The X axis is orthogonal to both of a first main surface **11***a* and a second main surface **11***b* of the heat-receiving block **11**, and the Y axis is orthogonal to both of the X and Z axes.

(19) The components of the cooling device **1** having the above-mentioned configuration are described, focusing on an example in which the cooling device **1** includes four primary pipe portions **12***a*, each of which is communicating with four secondary pipe portions **12***b*.

(20) As illustrated in FIG. **2**, the first main surface **11***a* of the heat-receiving block **11** is provided with heat emitters **31** fixed thereto. The heat emitters **31** include electronic components that emit heat during energization. The second main surface **11***b* of the heat-receiving block **11**, opposite to the first main surface **11***a*, is provided with grooves **11***c* extending in the Y-axis direction and grooves **11***d* extending in the Y-axis direction. Each of the grooves **11***c* receives the primary pipe portion **12***a* inserted thereto. The primary pipe portions **12***a* are fixed at the heat-receiving block **11** by any fixing procedure, such as bonding with an adhesive or soldering.

(21) Each of the grooves **11***d* receives the second heat pipe **13** inserted thereto. The second heat pipes **13** are fixed at the heat-receiving block **11** by any fixing procedure, such as bonding with an adhesive or soldering. The grooves **11***d* are located adjacent to the grooves **11***c*. Specifically, the grooves **11***d* are provided in the vicinity of the grooves **11***c* such that the frozen first refrigerant **15** can be melted by transfer of the heat from the second heat pipes **13** fitted in the grooves **11***d* to the primary pipe portions **12***a* fitted in the grooves **11***c*, as described below. The heat-receiving block **11** is made of a material having a high thermal conductivity, for example, a metal, such as copper or aluminum.

(22) Each of the first heat pipes **12** includes the primary pipe portion **12***a* and the secondary pipe portions **12***b* that are communicating with the primary pipe portion **12***a*. The first heat pipes **12** accommodate the first refrigerant **15**.

(23) The primary pipe portions **12***a* are fitted in the respective grooves **11***c* and fixed at the heat-receiving block **11**. The primary pipe portions **12***a* are fixed at the heat-receiving block **11** while being partially exposed from the heat-receiving block **11**. The primary pipe portions **12***a* are made of a material having a high thermal conductivity, for example, a metal, such as copper or aluminum.

(24) The secondary pipe portions **12***b* are fixed at the respective primary pipe portions **12***a* by a procedure, such as welding or soldering, and are communicating with the respective primary pipe portions **12***a*. The secondary pipe portions **12***b* extend in the direction away from the second main surface **11***b*. The secondary pipe portions **12***b* are made of a material having a high thermal conductivity, for example, a metal, such as copper or aluminum.

(25) Each of the second heat pipes **13** includes an end fitted in the groove **11***d* and fixed at the heat-receiving block **11**. The second heat pipe **13** defines a flow path of which the initial point and the terminal point coincide with each other and which winds between the edge close to the heat-receiving block **11** and the edge distant from the heat-receiving block **11**. In the section orthogonal to the Y axis, that is, in the XZ plane, the second heat pipe **13** fixed at the heat-receiving block **11** extends along parts of the outer peripheries of the primary pipe portions **12***a*. Specifically, as illustrated in FIG. **3**, the second heat pipe **13** of a self-excited oscillation type has a winding flow path of which the initial point and the terminal point coincide with each other. The second heat pipe **13** is bent at 90° along the bending line L**1** represented in the dashed and single-dotted line, fitted

in the groove **11***d*, and then fixed at the heat-receiving block **11**.

(26) As illustrated in FIG. **2**, the second heat pipe **13** includes a surface **13***a* that faces the positive Z-axis direction and a surface **13***b* that faces the negative X-axis direction. The second heat pipe **13** is located adjacent to the first heat pipes **12**. Specifically, the second heat pipe **13** is located adjacent to the primary pipe portions **12***a* such that the frozen first refrigerant **15** can be melted by transfer of the heat from the heat emitters **31** via the individual surfaces **13***a* and **13***b* to the primary pipe portions **12***a*. For example, the second heat pipe **13** is distant from the primary pipe portions **12***a* by 100 millimeters or less. The surfaces **13***a* and **13***b* may preferably be in contact with the primary pipe portions **12***a*.

(27) Each of the fins **14** has through holes, and is fixed across the secondary pipe portions **12***b* while the secondary pipe portions **12***b* extend through the respective through holes. The fins **14** can contribute to improve the cooling efficiency of the cooling device **1**.

(28) The first refrigerant **15** is enclosed in the first heat pipes **12** in the gas-liquid two-phase state. The first refrigerant **15** is made of a substance, such as water, evaporated when receiving heat from the heat emitters **31** and condensed into liquid when discharging heat to the air around the cooling device **1**.

(29) The second refrigerant **16** is enclosed in the second heat pipes **13** in the gas-liquid two-phase state. The internal space of each second heat pipe **13** is closed by droplets of the second refrigerant **16** due to a function of the surface tension of the second refrigerant **16**, so that the second refrigerant **16** in the liquid state and the second refrigerant **16** in the gas state are distributed inside the second heat pipe **13**. The second refrigerant **16** is made of a substance, such as water, evaporated when receiving heat from the heat emitters **31** and condensed into liquid when discharging heat to the air around the cooling device **1**.

(30) At normal temperature, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the second heat pipe **13** is higher than the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12**. The second heat pipes **13** therefore less readily freeze than the first heat pipes **12**. For example, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the second heat pipe **13** is preferably be 50%, while the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12** is preferably be 20%.

(31) The cooling device **1** having the above-mentioned configuration is installed in a power conversion apparatus **30**, as illustrated in FIGS. **4** and **5**. FIG. **5** is a sectional view taken along the line B-B of FIG. **4**. The power conversion apparatus **30** includes a housing **32**, the heat emitters **31** accommodated in the housing **32**, and the cooling device **1** to cool the heat emitters **31**. The housing **32** includes a partition **33** to divide the internal space of the housing **32** into a closed compartment **32***a* and an open compartment **32***b*. The closed compartment **32***a* accommodates the heat emitters **31**. The open compartment **32***b* accommodates the cooling device **1**. The partition **33** has an opening **33***a*. The opening **33***a* is closed by the first main surface **11***a* of the heat-receiving block **11** included in the cooling device **1**. The heat emitters **31** are mounted on the first main surface **11***a* that closes the opening **33***a*. The closing of the opening **33***a* by the first main surface **11***a* can prevent the external air, water, dust, or the like from entering the closed compartment **32***a*.

(32) The housing **32** also has intake and exhaust ports **34** in the two walls that face the open compartment **32***b* and are orthogonal to the Y-axis direction. Cooling air introduced through one of the intake and exhaust ports **34** in one wall flows between the secondary pipe portions **12***b* along the fins **14** and then exits through the other one of the intake and exhaust ports **34** in the other wall. The cooling device **1** transfers the heat from the heat emitters **31** to the cooling air, to thereby cool the heat emitters **31**.

(33) A mechanism of cooling the heat emitters **31** in the cooling device **1** having the above-mentioned configuration is described. When the heat emitters **31** emit heat, the heat is transferred from the heat emitters **31** via the heat-receiving block **11** and the primary pipe portions **12***a* to the

first refrigerant **15**. The transferred heat raises the temperature of the first refrigerant **15** and evaporates a part of the first refrigerant **15**. The evaporated first refrigerant **15** flows from the primary pipe portions **12***a* into the secondary pipe portions **12***b*, and travels inside the secondary pipe portions **12***b* toward the upper edges in the vertical direction of the secondary pipe portions **12***b*. During the travel inside the secondary pipe portions **12***b* toward the upper edges in the vertical direction of the secondary pipe portions **12***b*, the first refrigerant **15** discharges heat to the air around the cooling device **1** via the secondary pipe portions **12***b* and the fins **14**. The heat discharge from the first refrigerant **15** lowers the temperature of the first refrigerant **15**. The first refrigerant **15** is accordingly condensed into liquid. The first refrigerant **15** in the liquid state moves along the inner walls of the secondary pipe portions **12***b* and returns to the primary pipe portions **12***a*. When the first refrigerant **15** in the liquid state receives the heat from the heat emitters **31** via the heat-receiving block **11**, the first refrigerant **15** is evaporated again, flows into the secondary pipe portions **12***b*, and then travels toward the upper edges in the vertical direction of the secondary pipe portions **12***b*. The first refrigerant **15** thus repeats the above-described evaporation and condensation and thereby circulates, so that the heat generated at the heat emitters **31** is discharged to the air around the cooling device **1**, specifically, the air around the secondary pipe portions **12***b* and the fins **14**, resulting in cooling of the heat emitters **31**.

(34) If the heat emitted from the heat emitters **31** is transferred from the heat emitters **31** via the heat-receiving block **11** and the primary pipe portions **12***a* to the first refrigerant **15**, then the first refrigerant **15** that has not been evaporated, that is, the first refrigerant **15** in the liquid state has internal temperature differences and generates convection. The convection allows the first refrigerant **15** to diffuse and transfer the heat from the heat emitters **31** in the Y-axis direction, leading to efficient cooling of the heat emitters **31**.

(35) When the heat emitters **31** emit heat, the heat is also transferred from the heat emitters **31** via the heat-receiving block **11** and the second heat pipes **13** to the second refrigerant **16**. The transferred heat evaporates a part of the second refrigerant **16** in the liquid state. The evaporated second refrigerant **16** has an increased volume and urges the second refrigerant **16** in the liquid state and the second refrigerant **16** in the gas state to travel toward the edges distant from the heat-receiving block **11**, in other words, the upper edges in the vertical direction. During the travel inside the second heat pipes **13** toward the upper edges in the vertical direction, the evaporated second refrigerant **16** discharges heat to the air around the cooling device **1** via the second heat pipes **13**. The heat discharge from the second refrigerant **16** lowers the temperature of the second refrigerant **16**. The second refrigerant **16** is accordingly condensed into liquid. The second refrigerant **16** in the liquid state moves along the inner walls of the second heat pipes **13** downward in the vertical direction. When the second refrigerant **16** in the liquid state receives heat from the heat emitters **31** via the heat-receiving block **11**, the second refrigerant **16** is evaporated again. The second refrigerant **16** thus repeats the evaporation and condensation and thereby circulates, so that the heat generated at the heat emitters **31** is discharged to the air around the cooling device **1**, specifically, the air around the second heat pipes **13**, resulting in cooling of the heat emitters **31**.

(36) The frozen first refrigerant **15**, however, does not make the above-described circulation and convection, and does not enable the cooling device **1** to cool the heat emitters **31**. Specifically, if the air around the cooling device **1** reaches 0° C. or less during no energization of the electronic components included in the heat emitters **31**, the first refrigerant **15** made of water may freeze. The frozen first refrigerant **15** needs to be melted in order to prevent deterioration of the cooling efficiency of the cooling device **1**.

(37) A mechanism of melting the frozen first refrigerant **15** in the cooling device **1** is described. When the heat emitters **31** emit heat, the heat is transferred via the heat-receiving block **11** and the first heat pipes **12** to the first refrigerant **15**. The heat generated at the heat emitters **31** is also transferred to the second heat pipes **13**, and is then transferred from the individual surfaces **13***a* and **13***b* of the second heat pipes **13** located adjacent to the primary pipe portions **12***a*, via the primary

pipe portions **12***a* to the first refrigerant **15**. As a result, the heat is transferred to the frozen first refrigerant **15** in multiple ways, not only at the parts of the primary pipe portions **12***a* that face the heat-receiving block **11** but also at the parts of the primary pipe portions **12***a* that do not face the heat-receiving block **11** via the second heat pipes **13**. The cooling device **1** can therefore melt the frozen first refrigerant **15** more rapidly than an existing heat-pipe cooling device without the second heat pipes **13**.

(38) As described above, the cooling device **1** according to Embodiment 1 can rapidly melt the frozen first refrigerant **15** because of the second heat pipes **13**. The cooling device **1** can therefore cool the heat emitters **31** even in a low-temperature environment.

Embodiment 2

(39) The second heat pipes **13** may have any structure provided that the second heat pipes **13** less readily freeze than the first heat pipes **12** and can melt the frozen first refrigerant **15**. A cooling device **2** according to Embodiment 2 illustrated in FIGS. **6** and **7** includes second heat pipes **17** in place of the second heat pipes **13**. The cooling device **2** has the structure identical to that of the cooling device **1** except for the second heat pipes **17** and the shape of the heat-receiving block **11**. The cooling device **2** can be installed in the power conversion apparatus **30**, like the cooling device **1**.

(40) The heat-receiving block **11** included in the cooling device **2** is provided with grooves **11***e* instead of the grooves **11***d*. Each of the grooves **11***e* receives one end of the second heat pipe **17** inserted thereto.

(41) The one end of the second heat pipe **17** is fitted in the groove **11***e* and fixed at the heat-receiving block **11** by a procedure, such as bonding with an adhesive or soldering. The second heat pipe **17** includes a plate member **19** having an internal flow path **18**. Specifically, as illustrated in FIG. **8**, the plate member **19** has a flat shape and provided with the winding internal flow path **18** of which the initial point and the terminal point coincide with each other. This plate member **19** is bent at 90° along the bending line L**2** represented in the dashed and single-dotted line, thereby yielding the second heat pipe **17**. The flow path **18** accommodates the second refrigerant **16** in the gas-liquid two-phase state, as in Embodiment 1. The plate member **19** is made of a material excellent in thermal conductivity and processability, for example, a metal, such as copper or aluminum.

(42) The second heat pipe **17** fabricated by bending the plate member **19** as described above includes a surface **17***a* that faces the positive Z-axis direction and a surface **17***b* that faces the negative X-axis direction, as illustrated in FIG. **7**. The second heat pipe **17** is located adjacent to the first heat pipes **12**. Specifically, the second heat pipe **17** is located adjacent to the primary pipe portions **12***a* such that the frozen first refrigerant **15** can be melted by transfer of the heat from the heat emitters **31** via the individual surfaces **17***a* and **17***b* to the primary pipe portions **12***a*. The surfaces **17***a* and **17***b* may preferably be in contact with the primary pipe portions **12***a*.

(43) At normal temperature, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the flow path **18** in the second heat pipe **17** is higher than the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12**, as in Embodiment 1. The second heat pipes **17** therefore less readily freeze than the first heat pipes **12**. For example, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the flow path **18** in the second heat pipe **17** is preferably be 50%, while the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12** is preferably be 20%.

(44) A mechanism of cooling the heat emitters **31** in the cooling device **2** having the above-mentioned configuration is described. The first heat pipes **12** cool the heat emitters **31** by the mechanism identical to that in Embodiment 1. The following description is thus directed to a mechanism of the second heat pipes **17** cooling the heat emitters **31**.

(45) When the heat emitters **31** emit heat, the heat is transferred from the heat emitters **31** via the

heat-receiving block **11** and the plate members **19** to the second refrigerant **16**. The transferred heat evaporates a part of the second refrigerant **16** in the liquid state. The evaporated second refrigerant **16** has an increased volume and urges the second refrigerant **16** in the liquid state and the second refrigerant **16** in the gas state to travel toward the edges distant from the heat-receiving block **11**, in other words, the upper edges in the vertical direction. During the travel through the flow paths **18** toward the upper edges in the vertical direction, the evaporated second refrigerant **16** discharges heat to the air around the cooling device **2** via the plate members **19**. The heat discharge from the second refrigerant **16** lowers the temperature of the second refrigerant **16**. The second refrigerant **16** is accordingly condensed into liquid. The second refrigerant **16** in the liquid state moves along the inner walls of the flow paths **18** downward in the vertical direction. When the second refrigerant **16** in the liquid state receives heat from the heat emitters **31** via the heat-receiving block **11** and the plate members **19**, the second refrigerant **16** is evaporated again. The second refrigerant **16** thus repeats the evaporation and condensation and thereby circulates, so that the heat generated at the heat emitters **31** is discharged to the air around the cooling device **2**, specifically, the air around the second heat pipes **17**, resulting in cooling of the heat emitters **31**.

(46) In addition, a part of the heat transferred from the heat emitters **31** via the heat-receiving block **11** to the plate members **19** constituting the second heat pipes **17** is discharged from the plate members **19** directly to the ambient air, resulting in cooling of the heat emitters **31**.

(47) A mechanism of melting the frozen first refrigerant **15** in the cooling device **2** is described. When the heat emitters **31** emit heat, the heat is transferred via the heat-receiving block **11** and the first heat pipes **12** to the first refrigerant **15**, as in Embodiment 1.

(48) The heat generated at the heat emitters **31** is also transferred to the second heat pipes **17**, and is then transferred from the individual surfaces **17***a* and **17***b* of the second heat pipes **17** located adjacent to the primary pipe portions **12***a*, via the primary pipe portions **12***a* to the first refrigerant **15**. That is, the heat is transferred to the frozen first refrigerant **15** in multiple ways, not only at the parts of the primary pipe portions **12***a* that face the heat-receiving block **11** but also at the parts of the primary pipe portions **12***a* that do not face the heat-receiving block **11** via the second heat pipes **17**. The cooling device **2** can therefore melt the frozen first refrigerant **15** more rapidly than an existing heat-pipe cooling device without the second heat pipes **17**.

(49) As described above, the cooling device **2** according to Embodiment 2 can rapidly melt the frozen first refrigerant **15** because of the second heat pipes **17**. The cooling device **2** can therefore cool the heat emitters **31** even in a low-temperature environment.

(50) In addition, the flow paths **18** in the second heat pipes **17** are formed inside the plate members **19**, and are therefore less susceptible to a variation in the temperature of the air around the cooling device **2** and less readily freeze than the second heat pipes **13** included in the cooling device **1** according to Embodiment 1.

(51) Furthermore, the distance between the second heat pipes **17** and the heat emitters **31** is shorter than the distance between the primary pipe portions **12***a* and the heat emitters **31**. The heat generated at the heat emitters **31** is therefore transferred to the second heat pipes **17** more rapidly than to the primary pipe portions **12***a*. This configuration can achieve efficient heat transfer from the second heat pipes **17** via the primary pipe portions **12***a* to the first refrigerant **15**, leading to rapid melting of the frozen first refrigerant **15**.

Embodiment 3

(52) The second heat pipes **13** may have any structure provided that the second heat pipes **13** less readily freeze than the first heat pipes **12** and can melt the frozen first refrigerant **15**. A cooling device **3** according to Embodiment 3 illustrated in FIG. **9** includes second heat pipes **20** in place of the second heat pipes **13**. The cooling device **3** has the structure identical to that of the cooling device **1** except for the second heat pipes **20** and the shape of the heat-receiving block **11**. The cooling device **3** can be installed in the power conversion apparatus **30**, like the cooling devices **1** and **2**.

(53) The heat-receiving block **11** included in the cooling device **3** is provided with the grooves **11***e* like those of the heat-receiving block **11** included in the cooling device **2**, and grooves **11***f*. Each of the grooves **11***e* receives one end of the second heat pipe **20**, while the groove **11***f*, disposed above the groove **11***e* in the vertical direction with the two grooves **11***c* between the grooves **11***f* and **11***e*, receives the other end of the second heat pipe **20**.
(54) The one end of the second heat pipe **20** is fitted in the groove **11***e* while the other end is fitted in the groove **11***f*. The second heat pipe **20** is then fixed at the heat-receiving block **11** by a procedure, such as bonding with an adhesive or soldering. The second heat pipe **20** is fabricated by bending the plate member **19** like that in Embodiment 2. Specifically, as illustrated in FIG. **10**, the plate member **19** has a flat shape and provided with the winding internal flow path **18** of which the initial point and the terminal point coincide with each other. This plate member **19** is bent at 90° along the bending lines L**3** and L**4** represented in the dashed and single-dotted lines, thereby yielding the second heat pipe **20**. The direction of bending the plate member **19** along the bending line L**3** is identical to the direction of bending the plate member **19** along the bending line L**4**.
(55) The second heat pipe **20** fabricated by bending the plate member **19** as described above includes a surface **20***a* that faces the positive Z-axis direction, a surface **20***b* that faces the negative X-axis direction, and a surface **20***c* that faces the negative Z-axis direction, as illustrated in FIG. **9**. The second heat pipe **20** is located adjacent to the first heat pipes **12**. Specifically, the second heat pipe **20** is located adjacent to the primary pipe portions **12***a* such that the heat from the heat emitters **31** can be transferred via the individual surfaces **20***a*, **20***b*, and **20***c* to the primary pipe portions **12***a*. The surfaces **20***a*, **20***b*, and **20***c* of the second heat pipes **20** may preferably be in contact with the primary pipe portions **12***a*.
(56) At normal temperature, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the flow path **18** in the second heat pipe **20** is higher than the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12**, as in Embodiment 1. The second heat pipes **20** therefore less readily freeze than the first heat pipes **12**. For example, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the flow path **18** in the second heat pipe **20** is preferably be 50%, while the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12** is preferably be 20%.
(57) The first heat pipes **12** cool the heat emitters **31** by the mechanism identical to that in Embodiment 1. The second heat pipes **20** cool the heat emitters **31** by the mechanism identical to that in Embodiment 2. Both edges of each plate member **19** constituting the second heat pipe **20** are fixed at the heat-receiving block **11**, so that both edges of the flow path **18** are warmed by the heat transferred from the heat emitters **31** via the heat-receiving block **11**. Accordingly, the evaporated second refrigerant **16** has an increased volume and urges the second refrigerant **16** in the liquid state and the second refrigerant **16** in the gas state to travel toward the middle portion of the flow path **18**, in other words, the center of the plate member **19** in the longitudinal direction.
(58) A mechanism of melting the frozen first refrigerant **15** in the cooling device **3** is described. When the heat emitters **31** emit heat, the heat is transferred via the heat-receiving block **11** and the first heat pipes **12** to the first refrigerant **15**.
(59) The heat generated at the heat emitters **31** is also transferred to the second heat pipes **20**, and is then transferred from the individual surfaces **20***a*, **20***b*, and **20***c* of the second heat pipes **20** located adjacent to the primary pipe portions **12***a*, via the primary pipe portions **12***a* to the first refrigerant **15**. That is, the heat is transferred to the frozen first refrigerant **15** in multiple ways, not only at the parts of the primary pipe portions **12***a* that face the heat-receiving block **11** but also at the parts of the primary pipe portions **12***a* that do not face the heat-receiving block **11** via the second heat pipes **20**. The cooling device **3** can therefore melt the frozen first refrigerant **15** more rapidly than an existing heat-pipe cooling device without the second heat pipes **20**.
(60) In addition, the heat generated at the heat emitters **31** is transferred via both edges of the

second heat pipes **20** to the second heat pipes **20** because the edges of the second heat pipes **20** are fixed at the heat-receiving block **11**. The heat is therefore distributed to the entire second heat pipes **20** more rapidly in comparison to the cases of the second heat pipes **13** and **17**. The second heat pipes **20** can thus melt the frozen first refrigerant **15** more rapidly than the second heat pipes **13** and **17**.

(61) As described above, the cooling device **3** according to Embodiment 3 can rapidly melt the frozen first refrigerant **15**, because of the second heat pipes **20** causing the heat from the heat emitters **31** to be transferred from the surfaces **20***a*, **20***b*, and **20***c* via the primary pipe portions **12***a* to the first refrigerant **15**. The cooling device **3** can therefore cool the heat emitters **31** even in a low-temperature environment.

Embodiment 4

(62) Although the second heat pipes **13**, **17**, and **20** according to Embodiments 1 to 3 extend along parts of the outer peripheries of the primary pipe portions **12***a* in the XZ plane, the second heat pipes **13**, **17**, and **20** may extend along parts of the outer peripheries of the primary pipe portions **12***a* in the XZ plane and also extend along the secondary pipe portions **12***b*. As illustrated in FIG. **11**, a cooling device **4** according to Embodiment 4 includes second heat pipes **21** in place of the second heat pipes **13**. The cooling device **4** has the structure identical to that of the cooling device **1** except for the second heat pipes **21** and the shape of the heat-receiving block **11**. The cooling device **4** can be installed in the power conversion apparatus **30**, like the cooling devices **1** to **3**.

(63) The heat-receiving block **11** included in the cooling device **4** is provided with grooves **11***g* instead of the grooves **11***d*. Each of the grooves **11***g* receives the second heat pipe **21** inserted therein. The groove **11***g* is located adjacent to the groove **11***c* at the position above the groove **11***c* in the vertical direction. Specifically, the groove **11***g* is located adjacent to the groove **11***c*, such that the frozen first refrigerant **15** can be melted by transfer of the heat from the second heat pipe **21** fitted in the groove **11***g* to the primary pipe portion **12***a* fitted in the groove **11***c*.

(64) The second heat pipe **21** is fitted in the groove **11***g* and then fixed at the heat-receiving block **11** by a procedure, such as bonding with an adhesive or soldering. The second heat pipe **21** includes the internal flow path **18**. Specifically, as illustrated in FIG. **12**, the plate member **19** has a flat shape and provided with the winding internal flow path **18** of which the initial point and the terminal point coincide with each other. This plate member **19** is bent at 90° along the bending line L**5** represented in a dashed and single-dotted line and bent along the bending line L**6** represented in a dashed and single-dotted line at the angle defined between the horizontal direction and the direction of extension of the secondary pipe portion **12***b*, thereby yielding the second heat pipe **21**. The direction of bending the plate member **19** along the bending line L**5** is opposite to the direction of bending the plate member **19** along the bending line L**6**. The second heat pipe **21** is fabricated by bending the plate member **19** at the angle defined between the horizontal direction and the direction of extension of the secondary pipe portion **12***b*, and thus extends along the secondary pipe portion **12***b* while being fixed at the heat-receiving block **11**.

(65) The second heat pipe **21** fabricated by bending the plate member **19** as described above includes a surface **21***a* that faces the negative Z-axis direction, a surface **21***b* that faces the negative X-axis direction, and a surface **21***c* extending along the secondary pipe portion **12***b*, as illustrated in FIG. **11**. The second heat pipe **21** is located adjacent to the primary pipe portion **12***a*. Specifically, the second heat pipe **21** is located adjacent to the primary pipe portion **12***a*, such that the heat from the heat emitters **31** can be transferred via the individual surfaces **21***a* and **21***b* to the primary pipe portion **12***a*. The surfaces **21***a* and **21***b* of the second heat pipe **21** may preferably be in contact with the primary pipe portion **12***a*. Also, the second heat pipe **21** is located adjacent to the secondary pipe portion **12***b* and extends along the secondary pipe portion **12***b* such that the heat from the heat emitters **31** can be transferred via the surface **21***c* to the secondary pipe portion **12***b*. The surface **21***c* of the second heat pipe **21** may preferably be in contact with the secondary pipe portion **12***b*.

(66) At normal temperature, the ratio of the amount of the second refrigerant **16** in the liquid state

to the volume of the flow path **18** in the second heat pipe **21** is higher than the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12**, as in Embodiment 1. The second heat pipes **21** therefore less readily freeze than the first heat pipes **12**. For example, the ratio of the amount of the second refrigerant **16** in the liquid state to the volume of the flow path **18** in the second heat pipe **21** is preferably be 50%, while the ratio of the amount of the first refrigerant **15** in the liquid state to the volume of the first heat pipe **12** is preferably be 20%.
(67) The first heat pipes **12** cool the heat emitters **31** by the mechanism identical to that in Embodiment 1. The second heat pipes **21** cool the heat emitters **31** by the mechanism identical to that in Embodiment 2.
(68) A mechanism of melting the frozen first refrigerant **15** in the cooling device **4** is described. When the heat emitters **31** emit heat, the heat is transferred via the heat-receiving block **11** and the first heat pipes **12** to the first refrigerant **15**.
(69) The heat generated at the heat emitters **31** is also transferred to the second heat pipes **21**, and is then transferred from the individual surfaces **21***a* and **21***b* of the second heat pipes **21** located adjacent to the primary pipe portions **12***a*, via the primary pipe portions **12***a* to the first refrigerant **15**. That is, the heat is transferred to the frozen first refrigerant **15** in multiple ways, not only at the parts of the primary pipe portions **12***a* that face the heat-receiving block **11** but also at the parts of the primary pipe portions **12***a* that do not face the heat-receiving block **11** via the second heat pipes **21**.
(70) In addition, the heat transferred to the second heat pipes **21** is transferred from the surfaces **21***c* via the secondary pipe portions **12***b* to the first refrigerant **15**. The heat is thus also transferred to the first refrigerant **15** frozen in the secondary pipe portions **12***b*. The cooling device **4** can therefore melt the frozen first refrigerant **15** more rapidly than an existing heat-pipe cooling device without the second heat pipes **21**. Furthermore, the cooling device **4** can rapidly melt the first refrigerant **15** frozen in the secondary pipe portions **12***b*.
(71) As described above, the cooling device **4** according to Embodiment 4 can rapidly melt the frozen first refrigerant **15**, because of the second heat pipes **21** causing the heat from the heat emitters **31** to be transferred from the individual surfaces **21***a* and **21***b* via the primary pipe portions **12***a* to the first refrigerant **15** and causing the heat to be transferred from the surfaces **21***c* via the secondary pipe portions **12***b* to the first refrigerant **15**. The cooling device **4** can therefore cool the heat emitters **31** even in a low-temperature environment.
(72) The above-described embodiments are not construed as limiting the present disclosure. For example, some of the above-described embodiments may be arbitrarily combined. Specifically, the second heat pipes **13** included in the cooling device **1** may have the shape identical to that of the second heat pipes **20** or **21**.
(73) The heat-receiving block **11** does not necessarily have a plate shape, and may have any shape provided that the heat emitters **31** can be fixed to the first main surface **11***a* and the first heat pipes **12** can be fixed at the heat-receiving block **11**.
(74) The first heat pipe **12** may have any structure and shape provided that the first heat pipe **12** can discharge the heat transferred from the heat emitters **31**. For example, the first heat pipe **12** may include the primary pipe portion **12***a* alone. The section of the primary pipe portion **12***a* orthogonal to the longitudinal direction does not necessarily have a circular shape and may also have a flattened shape. Also, the section of the secondary pipe portion **12***b* orthogonal to the longitudinal direction does not necessarily have a circular shape and may also have a flattened shape. The flattened shape indicates a shape formed by narrowing the width of a part of the circular shape than the original width and encompasses elliptical, streamline, and elongated circular shapes. The elongated circular shape indicates a shape defined by circles having the same diameter and the straight lines connecting the contours of the circles with each other. In this case, when the primary pipe portion **12***a* is fixed at the heat-receiving block **11** such that the longer axis of the flattened

shape is in parallel to the Z-axis direction, this arrangement can improve the efficiency of heat transfer from the heat-receiving block **11** to the primary pipe portion **12***a*. Furthermore, when the secondary pipe portion **12***b* is fixed at the primary pipe portion **12***a* such that the longitudinal direction of the secondary pipe portion **12***b* coincides with the direction of flow of the cooling air, this configuration can reduce turbulence in the vicinity of the secondary pipe portion **12***b*, leading to improvement of the cooling efficiency.

(75) The second heat pipes **13**, **17**, **20**, and **21** may have any shape provided that these pipes can melt the frozen first refrigerant **15**. For example, the second heat pipes **17**, **20**, and **21** may include a member having any shape and including the internal flow path **18**.

(76) As another example, FIG. **13** illustrates second heat pipes **22** each of which includes a segment extending along parts of the outer peripheries of the primary pipe portions **12***a* in the XZ plane and another segment extending along the secondary pipe portion **12***b*. The second heat pipe **22** includes a bent segment **22***a* that extends along parts of the outer peripheries of the primary pipe portions **12***a* in the XZ plane like the second heat pipe **17**, and a linear segment **22***b* that extends from the second main surface **11***b* of the heat-receiving block **11** along the secondary pipe portion **12***b*.

(77) The flow path **18** may have any shape provided that the flow path **18** allows the second refrigerant **16** enclosed therein to circulate. For example, the flow path **18** may have a ring shape.

(78) The second heat pipe **13**, **17**, **20**, or **21** does not necessarily be a self-excited heat pipe and may also be a loop heat pipe **23** illustrated in FIG. **14**. The loop heat pipe **23** includes an evaporator **23***a* to transfer the heat generated at the heat emitters **31** to the second refrigerant **16** and thereby evaporate the second refrigerant **16**, a vapor pipe **23***b* through which the evaporated second refrigerant **16** flows, a condenser **23***c* to discharge the heat transferred from the second refrigerant **16** and thereby condense the second refrigerant **16** into liquid, a liquid pipe **23***d* through which the second refrigerant **16** in the liquid state flows, and a reservoir **23***e* to retain a part of the second refrigerant **16** flowing through the liquid pipe **23***d* and thereby adjust the amount of the second refrigerant **16** flowing from the liquid pipe **23***d* into the evaporator **23***a*. In this case, the evaporator **23***a* is fixed at the heat-receiving block **11**.

(79) Alternatively, the loop heat pipe **23** may be formed inside a plate member, like the second heat pipes **17**, **20**, and **21**. In this case, the plate member including the internal loop heat pipe **23** is fixed at the heat-receiving block **11** such that the evaporator **23***a* is located adjacent to the heat-receiving block **11**.

(80) The heat emitters **31** fixed at the heat-receiving block **11** may be switching elements including wide bandgap semiconductors. The wide bandgap semiconductor contains, for example, a silicon carbide, gallium nitride martial, or diamond. The switching element including a wide bandgap semiconductor has a reduced size in comparison to a switching element including silicon, and therefore generates a larger amount of heat per unit area. The heat generated in the wide bandgap semiconductors is received at the second heat pipes **13**, **17**, **20**, **21**, or **22**, which less readily freeze than the first heat pipes **12**, so that the second heat pipes **13**, **17**, **20**, **21**, or **22** can rapidly melt the frozen first refrigerant **15**.

(81) The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full range of equivalents to which such claims are entitled.

REFERENCE SIGNS LIST

(82) **1**, **2**, **3**, **4** Cooling device **11** Heat-receiving block **11***a* First main surface **11***b* Second main surface **11***c*, **11***d*, **11***e*, **11***f*, **11***g* Groove **12** First heat pipe **12***a* Primary pipe portion **12***b* Secondary pipe portion **13**, **17**, **20**, **21**, **22** Second heat pipe **13***a*, **13***b*, **17***a*, **17***b*, **20***a*, **20***b*, **20***c*, **21***a*, **21***b*, **21***c*

Surface **14** Fin **15** First refrigerant **16** Second refrigerant **18** Flow path **19** Plate member **22***a* Bent segment **22***b* Linear segment **23** Loop heat pipe **23***a* Evaporator **23***b* Vapor pipe **23***c* Condenser **23***d* Liquid pipe **23***e* Reservoir **30** Power conversion apparatus **31** Heat emitter **32** Housing **32***a* Closed compartment **32***b* Open compartment **33** Partition **33***a* Opening **34** Intake and exhaust port L**1**, L**2**, L**3**, L**4**, L**5**, L**6** Bending line

Claims

1. A cooling device comprising: a heat-receiving block comprising a first main surface to which a heat emitter is fixed; a first heat pipe fixed at the heat-receiving block; a second heat pipe fixed at the heat-receiving block and located adjacent to the first heat pipe; a first refrigerant in a gas-liquid two-phase state enclosed in the first heat pipe; and a second refrigerant in a gas-liquid two-phase state enclosed in the second heat pipe, wherein a ratio of an amount of the second refrigerant in a liquid state to a volume of the second heat pipe is higher than a ratio of an amount of the first refrigerant in a liquid state to a volume of the first heat pipe, at normal temperature, and the second heat pipe defines a flow path winding between a first edge and a second edge opposite the first edge, the first edge being closer to the heat-receiving block and closer to the first heat pipe than the second edge, the flow path having an initial point and a terminal point that coincide with each other on the first edge, the flow path extending from the first edge to the second edge, turning around and extending to the first edge, turning around and extending to the second edge again, then turning around and extending to the first edge again, and at least part of the second heat pipe is located adjacent to a portion of the first heat pipe that is fixed at the heat-receiving block.

2. The cooling device according to claim 1, wherein the second heat pipe comprises a member comprising an internal flow path.